Reducing Scrap in Injection Molding:
A DAG-Informed Causal Decision Analysis

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Abstract

We analyze 5,000 half-hour production intervals from 12 injection-molding machines across four Karcher plants to identify actionable levers for reducing scrap rate. The mean scrap rate is 4.44%, and 78% of intervals exceed the 3.2% pass/fail threshold. The analysis is framed as an *intervention* problem — which settings should a plant manager change, in what direction, and by how much — rather than as a prediction problem.

The analysis is structured by a Directed Acyclic Graph (DAG) supplied with the challenge ontology. Under the DAG-specified adjustment strategy, the central finding is a sign reversal for the most important lever: the raw Pearson correlation between cooling time and scrap rate is $\rho = +0.28$, while the adjusted standardized coefficient is $\beta = -1.75$ (corresponding to approximately −0.41 percentage points of scrap per additional second of cooling). A predictive analysis acting on the raw correlation would likely recommend shortening cooling time; the DAG-adjusted estimate points in the opposite direction.

A coordinated five-action package — extending cooling time, capping mold temperature, adjusting dryer dewpoint at high humidity, reducing injection pressure on worn tooling, and shortening maintenance intervals — is estimated to reduce mean scrap from 4.44% to approximately 3.85% (−13% relative). All estimates are model-implied from observational data under the stated DAG assumptions. A two-week controlled pilot is recommended before plant-wide deployment.

Key Takeaways
→ Scrap is chronic, not anomalous: 78% of production intervals exceed the 3.2% pass/fail threshold.
→ Cooling time is the key finding: its raw correlation with scrap is positive (+0.28), but the DAG-adjusted estimate is protective ($\beta = -1.75$). Acting on the raw signal would likely increase scrap.
→ The five-action package is estimated to reduce mean scrap from 4.44% to ~3.85% (−13% relative) with a ~1% cycle-time cost and negligible energy impact. → All magnitudes are model-implied from observational data and are conditional on the DAG being correctly specified. → A two-week pilot at VN_QUANGNAM with defect-mix monitoring is required before plant-wide rollout.

1. Introduction

Injection molding scrap at the Karcher plants is chronic rather than anomalous. Across 5,000 production intervals spanning 75 days (January–March 2026) on 12 machines at four plants, the mean scrap rate is 4.44% and 78% of intervals fail the 3.2% threshold. The dominant defect is

warpage, accounting for 33% of all intervals; flash and dimensional deviation contribute a further 37%. The task is not to detect unusually bad intervals — most intervals are bad — but to lower the overall distribution.

This operational goal requires an **intervention** framing, not a predictive one. A model trained to forecast scrap identifies variables that co-vary with the outcome; it does not distinguish controllable levers from downstream symptoms, confounded proxies, or reverse-directed signals. A variable that forecasts scrap well may not reduce it when changed, and may not even have the right sign for intervention purposes. Cooling time illustrates this precisely: its raw positive correlation with scrap ($\rho = +0.28$) would motivate a recommendation to shorten cooling time, while DAG-adjusted analysis points in the opposite direction. The distinction between predicting and intervening is the central methodological motivation for this analysis.

The formal framework for estimating intervention effects from observational data is Pearl's do-calculus (Pearl, 2000; Pearl, Glymour & Jewell, 2016). The central object is a Directed Acyclic Graph encoding structural causal assumptions; identification results — particularly the backdoor criterion — specify which adjustment sets yield less-confounded effect estimates from observational data. The challenge organizers supply a DAG and ontology that serve as the structural prior. Applications in process manufacturing consistently find that correlational feature rankings misdirect interventions, and that DAG-adjusted estimates materially change the recommended intervention direction (Rendall et al., 2019; Vowels et al., 2022).

All causal effect estimates reported here are conditional on (a) the supplied DAG being correctly specified, (b) no material unmeasured confounding beyond what is encoded in the graph, (c) correct model specification, and (d) the observational nature of the data. These conditions are discussed in Section 6.

Section 2 presents the causal structure and identification strategy. Section 3 describes the estimation methods. Section 4 presents results, with the cooling sign change as the central finding. Section 5 translates findings into a five-action recommendation. Sections 6 and 7 address limitations and conclusions.

2. Causal Setup and DAG

2.1 Variable Classification

The ontology partitions all variables into five roles. This classification directly determines how each variable enters the estimation strategy — whether it is changed, adjusted for, or excluded.

- **Controllable process levers**: dryer dewpoint, barrel temperature, mold temperature, injection pressure, hold pressure, cooling time, screw speed, shot size. Real-time setpoints an operator can adjust.
- **Planning-level lever**: days since last maintenance. A scheduling-policy decision rather than a per-interval setpoint.
- **Context controls**: product variant, cavity count, part weight, and plant/machine/mold/resin-lot identifiers. Conditioned on in every model; no intervention is recommended on these variables.
- **Confounders**: ambient humidity, ambient temperature, and operator shift. These have causal paths to both process lever settings and the outcome; they are adjusted for in every regression to block backdoor paths.
- **Operator experience (covariate)**: adjusted for in all specifications, but its positive adjusted coefficient is not given a direct causal interpretation — see Section 4.4.

- **Mediators**: resin moisture, calibration drift, and tool wear. These transmit the effects of upstream causes toward the outcome. When estimating a lever's *total effect*, mediators are excluded from the adjustment set: conditioning on them blocks part of the causal pathway from the lever to scrap, yielding a controlled direct effect rather than the total effect of interest.
- **Outcomes**: scrap rate (primary KPI), defect type, pass/fail flag, parts produced, energy consumption, and cycle time. Excluded from all predictor sets; cycle time is used only as a trade-off outcome.

2.2 Causal Pathways

Figure 1 renders the causal graph. The DAG encodes five chains linking upstream drivers, through process levers and optionally mediators, to the scrap rate KPI.

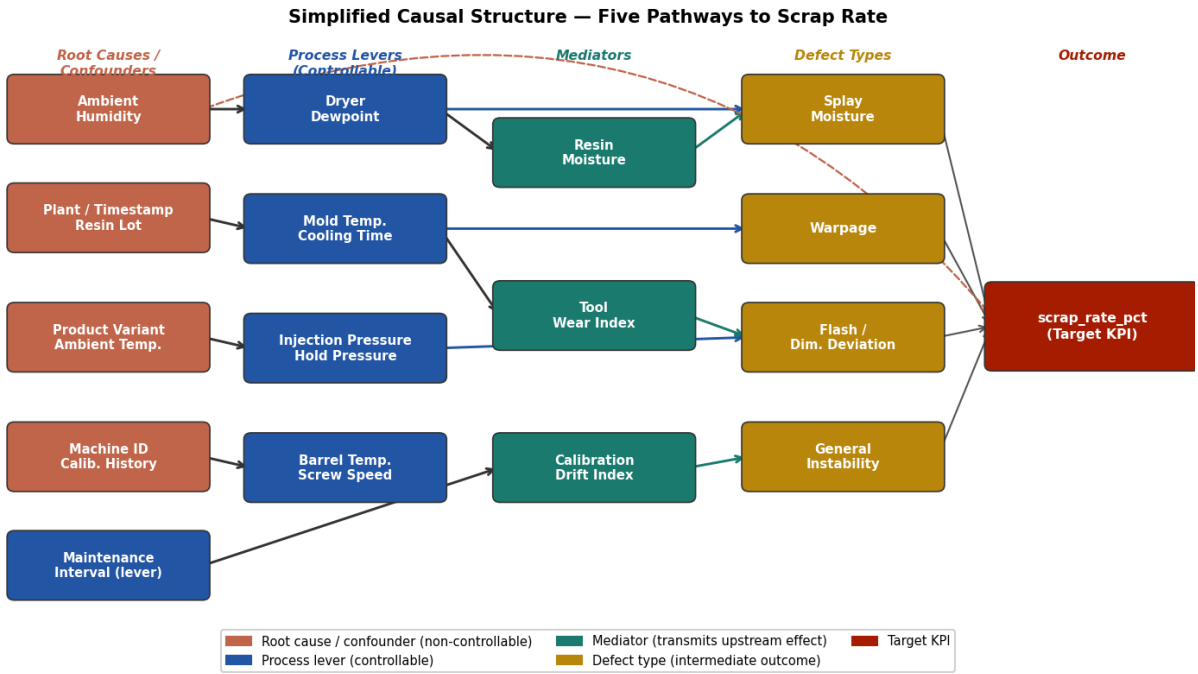

Figure 1 — Simplified causal structure used for adjustment. Environmental and product-context variables (coral) influence both process settings and scrap; mediators (teal) such as resin moisture, calibration drift, and tool wear transmit upstream effects. Process levers (blue) are the intervention targets. Adjustment sets are chosen to block backdoor paths without conditioning on mediators. Dashed arrow: direct confounding path from ambient humidity to scrap rate.

Warpage (33% of intervals). Mold temperature and cooling time jointly determine residual thermal stress at ejection. The interaction between them is the primary actionable signal: at mold temperatures above 75 °C, extending cooling time is estimated to produce a meaningful reduction in warpage-related scrap (Section 4.3). This is the pathway with the strongest evidence for intervention.

Flash (19% of intervals). Tool wear and injection pressure interact to create flash defects. The relevant signal is conditional on wear state: elevated pressure on heavily worn tooling is associated with substantially higher flash rates, while the same pressure on lightly worn tooling is not.

Dimensional deviation (18% of intervals). Inadequate hold pressure and tool wear jointly cause uneven shrinkage and imprecision. The adjusted effect of hold pressure is mildly protective ($\beta \approx -0.09$) but borderline significant; it is tracked but not included in the primary action package.

Splay moisture (14% of intervals). Atmospheric moisture absorbed by resin vaporizes during injection. The dryer dewpoint is the controllable lever, but the structural model linking dewpoint to resin moisture explains only 9% of moisture variance at the interval level ($R^2 = 0.09$), which bounds the expected intervention effect.

Process stability (cross-pathway). Deferred maintenance leads to calibration drift, which destabilizes process parameters across all defect modes simultaneously. The direct adjusted effect of maintenance interval is small, but the mechanism runs through calibration drift — a mediator that must not enter the adjustment set when estimating the maintenance lever's total effect.

2.3 Identification Strategy

For each lever T, the target estimand is the average causal effect:

$$ACE(T, t_1, t_0) = E[Y \mid do(T = t_1)] - E[Y \mid do(T = t_0)] \quad (1)$$

Under the **backdoor criterion** (Pearl, 2000, §3.3), this estimand is identified from observational data by conditioning on any set Z that blocks all non-causal (backdoor) paths from T to Y without containing any descendant of T:

$$E[Y \mid do(T = t)] = \Sigma_z \; E[Y \mid T = t, Z = z] \cdot P(Z = z) \quad (2)$$

For cooling time, the backdoor paths run through mold temperature (a common cause of operator cooling decisions and warpage risk), part weight and shot size (which set the thermal mass being cooled), and the environmental confounders. The adjustment set Z therefore includes those variables plus machine, mold, product-variant, and operator-shift fixed effects. Mediators — resin moisture, calibration drift, tool wear — are excluded from Z to preserve the total-effect estimand. Conditioning on them would block part of the causal pathway and yield only the controlled direct effect. Table 2A in Section 4.1 lists the adjustment set for each lever.

3. Method

3.1 Data

The dataset contains 5,000 rows × 33 columns. Each row is one 30-minute production interval per (machine × mold × product variant × resin lot), spanning 2026-01-05 to 2026-03-20 (75 days). No missing values; zero full-row duplicates. Minor data issues: 13 clamp-force values marginally above the 4,400 kN ceiling were clipped (sensor noise); 49 key-level duplicate pairs differing on product variant or resin lot are likely synthetic-data artefacts and were retained. The KPI is scrap_rate_pct as declared by the ontology (Pearson $\rho = 0.80$ with the independently computed ratio; max $|\Delta| = 7.4$ p.p.). Cycle time mechanically subsumes cooling time and is excluded from all predictor sets, used only as a trade-off outcome.

3.2 Estimation Stack

Four methods are applied in triangulation. Pearson correlation is used as a diagnostic rather than a decision rule, because it can reflect confounding or reverse causation. A recommendation is included in the action package when the DAG-adjusted estimate, counterfactual simulation, process theory, and relevant diagnostics support the same intervention direction.

(1) Pearson correlation — raw linear association. Used as a diagnostic screen; ρ can reflect confounding or reverse causation and is not treated as a causal estimate.

(2) Gradient-boosted regression (GBR) — nonlinear predictive benchmark and the world-approximator inside the counterfactual simulator:

$$\hat{y}(x) = \Sigma(m=1 \text{ to } M)\ \nu \cdot f_m(x) \quad (3)$$

Parameters: M = 400 trees, learning rate v = 0.05, depth = 3; 5-fold cross-validated $R^2 = 0.64$. Feature importance is reported as a predictive diagnostic only.

(3) Adjusted linear regression — primary causal effect estimator. For each candidate lever T:

$$Y_i = \beta_0 + \beta_1 T_i + \gamma^T Z_i + \alpha\text{machine} + \alpha\text{mold} + \alpha\text{variant} + \alpha\text{shift} + \varepsilon_i \quad (4)$$

All continuous features are z-scored so that β_1 is directly comparable across levers as a standardized effect size. The relationship between the standardized coefficient and the natural-unit coefficient is:

$$\beta_1(\text{std}) = \tilde{\beta_1} \cdot (\sigma T / \sigma Y) \quad (5)$$

For cooling time: the natural-unit coefficient is $\tilde{\beta}_1 = -0.41$ p.p./s; with σ(cooling) ≈ 4.3 s and σ(scrap) ≈ 1.0 p.p., this yields $\beta_1(\text{std}) \approx -1.75$. The standardized coefficient enables cross-lever comparison in Table 2A; the natural-unit coefficient is used for operational sizing and worked examples.

(4) Counterfactual simulation — quantifies expected intervention magnitude using the fitted GBR:

$$\hat{\Delta}_i = \hat{y}(do(T = t^*), x_i) - \hat{y}(do(T = t), x_i) \quad (6)$$

Each row is intervened on individually: T is shifted to the target value t*, all other features held at observed values, and the GBR is re-evaluated. Shifted values are clipped to ontology-declared physical ranges. For the dryer dewpoint lever, the shift is propagated through a structural moisture sub-model before entering ŷ, so that dewpoint is not treated as a direct cause of scrap. The population average treatment effect is the mean of $\hat{\Delta}_i$ across all n rows. This is observational simulation conditional on the GBR fit; it is not equivalent to a randomized experiment.

4. Results

4.1 Defect Burden and Adjusted Effect Estimates

Defect type	Count	Mean scrap (%)	Share of total
warpage	1,650	4.50	33.4%
flash	973	4.42	19.4%
dimensional deviation	936	4.30	18.1%
splay moisture	649	4.74	13.9%
sink mark	409	4.15	7.6%
short shot	300	4.45	6.0%

burn mark	78	4.38	1.5%
none	5	2.05	0.1%

Table 1 — Defect-type contribution to scrap burden across all 5,000 intervals.

Note that **defect burden** and **intervention leverage** are distinct quantities. Burden measures a defect's share of current scrap; leverage measures the expected change in scrap under a feasible setpoint shift. They diverge notably for the moisture pathway: splay moisture has the highest per-interval mean scrap (4.74%), but the dryer lever controls only about 9% of resin moisture variance, limiting its expected impact. Where burden and leverage agree — as with the warpage/cooling pathway — confidence in the recommendation is strongest.

Table 2A presents DAG-adjusted estimates for actionable process and scheduling levers. Table 2B presents diagnostic, mediator, and non-actionable variables for completeness. All continuous features are z-scored; coefficients are therefore directly comparable across rows.

Table 2A — Actionable levers (process setpoints and maintenance scheduling)

Lever	Adjustment set (beyond FE)	β (std)	95% CI	Per op. step
Cooling time	mold temp, part, shot, humidity, temp, maint	−1.75	[−1.85, −1.65]	−0.41 p.p. / +1 s
Mold temperature	cooling, barrel, part, confounders	+0.88	[+0.80, +0.96]	+0.40 p.p. / +5 °C
Injection pressure	wear, clamp, barrel, batch, part, hold, conf.	+0.31	[+0.22, +0.41]	+0.11 p.p. / +50 bar
Dryer dewpoint	humidity, temp	+0.09	[+0.05, +0.13]	+0.15 p.p. / +5 °C
Maintenance interval	humidity, temp	+0.10	[+0.07, +0.14]	+0.07 p.p. / +7 days
Hold pressure	injection, wear, confounders	−0.09	[−0.18, −0.00]	−0.05 p.p. / +50 bar
Barrel temperature	pressure, mold, batch, confounders	−0.13	[−0.19, −0.07]	−0.05 p.p. / +5 °C
Screw speed	shot, barrel, confounders	+0.01	[−0.03, +0.05]	≈ 0

Table 2A — DAG-adjusted effect estimates for actionable levers. Standardized β from z-scored features; 95% CIs from 300-replicate row-wise bootstrap. FE = machine/mold/variant/shift fixed effects. Positive β indicates the variable is associated with higher scrap under adjustment; negative β is associated with lower scrap. Per op. step: natural-unit effect via equation (5).

Table 2B — Diagnostic, mediator, and non-actionable variables (reported for completeness; not recommended for direct intervention)

Variable	Role / note	β (std)	95% CI	Interpretation
Ambient humidity	Confounder — not a lever	+0.42	[+0.37, +0.47]	Adjusted for; not actioned
Resin moisture	Mediator (dryer pathway)	+0.24	[+0.17, +0.29]	Reflects dryer + humidity effect
Tool wear index	Mediator (maintenance pathway)	+0.25	[+0.21, +0.28]	Reduced by maintenance cap
Calibration drift	Mediator (maintenance pathway)	+0.05	[−0.02, +0.13]	CI crosses zero; not actioned
Operator experience	Covariate (assignment bias likely)	+0.12	[+0.09, +0.16]	No causal interpretation; see §4.4

Table 2B — Non-actionable variables. Ambient humidity is a confounder adjusted for in all regressions; resin moisture, tool wear, and calibration drift are mediators that transmit upstream effects. The operator experience coefficient does not support a direct causal recommendation (see Section 4.4).

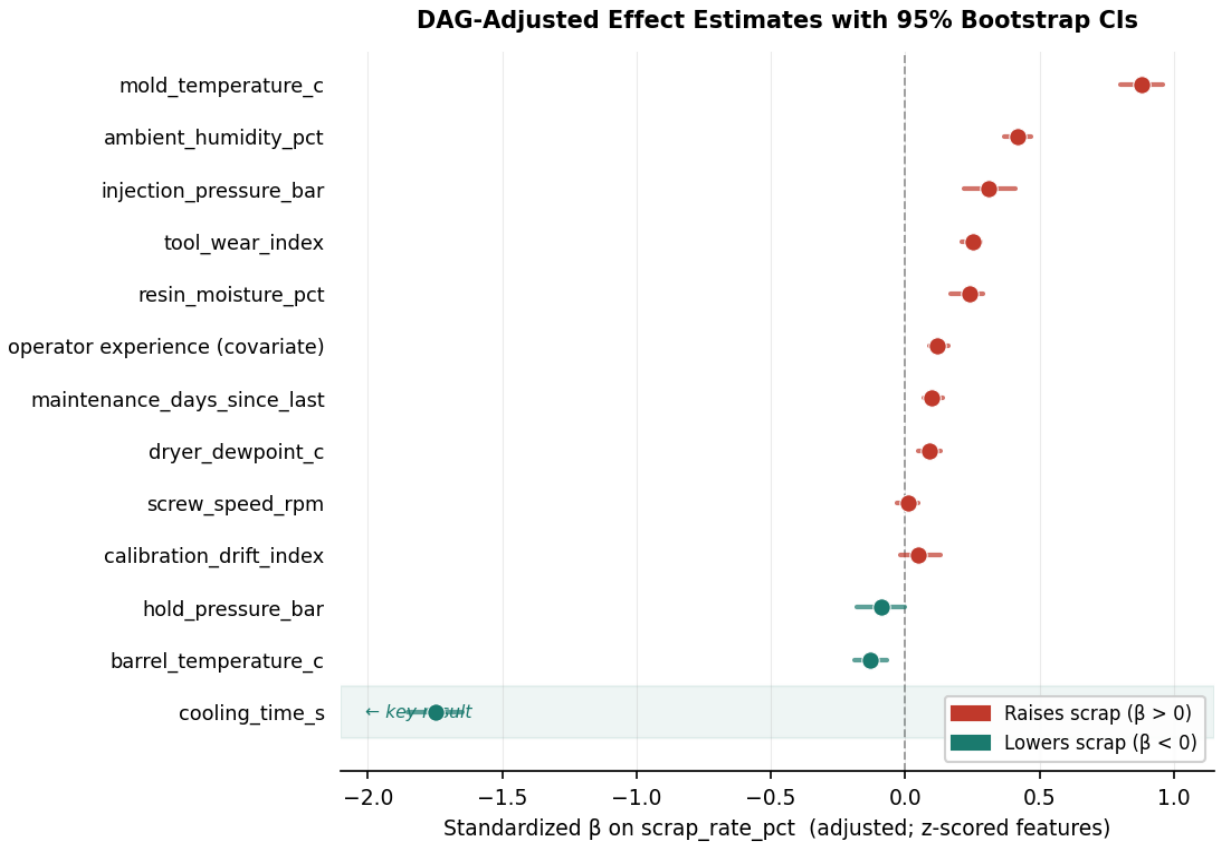

Figure 2 — DAG-adjusted standardized effect estimates (β) with 95% bootstrap CIs. Teal markers indicate an estimated protective association with scrap; red markers indicate an estimated positive association. The cooling-time estimate (highlighted band) is the only lever where sign reverses between raw Pearson correlation and adjusted estimate. The calibration drift CI crosses zero and is not actioned.

4.2 The Cooling-Time Sign Change

The raw Pearson correlation between cooling time and scrap rate is $\rho = +0.28$. A purely predictive pipeline acting on this value would rank cooling time as a positive predictor and likely recommend shortening cooling. The DAG-adjusted analysis points in the opposite direction.

After conditioning on mold temperature, part weight, and shot size — variables that plausibly drive operator decisions to extend cooling — the partial correlation reverses to $\rho = -0.37$. Under the full adjusted regression (equation 4), the standardized coefficient is $\beta = -1.75$, corresponding to −0.41 p.p. per additional second of cooling (natural-unit coefficient $\tilde{\beta} = -0.41$ p.p./s from equation 5).

The proposed mechanism is reverse causation. Operators already extend cooling time reactively in response to observed thermal stress in high-temperature molds. The raw positive correlation therefore reflects their residual compensation for a poor thermal state — it is a signal about where problems already exist, not about what caused them. Adjusting for the variables that create both the thermal stress and the operator response substantially reduces this confounding and reveals an estimated protective association. Figure 3 illustrates this shift from raw to adjusted association.

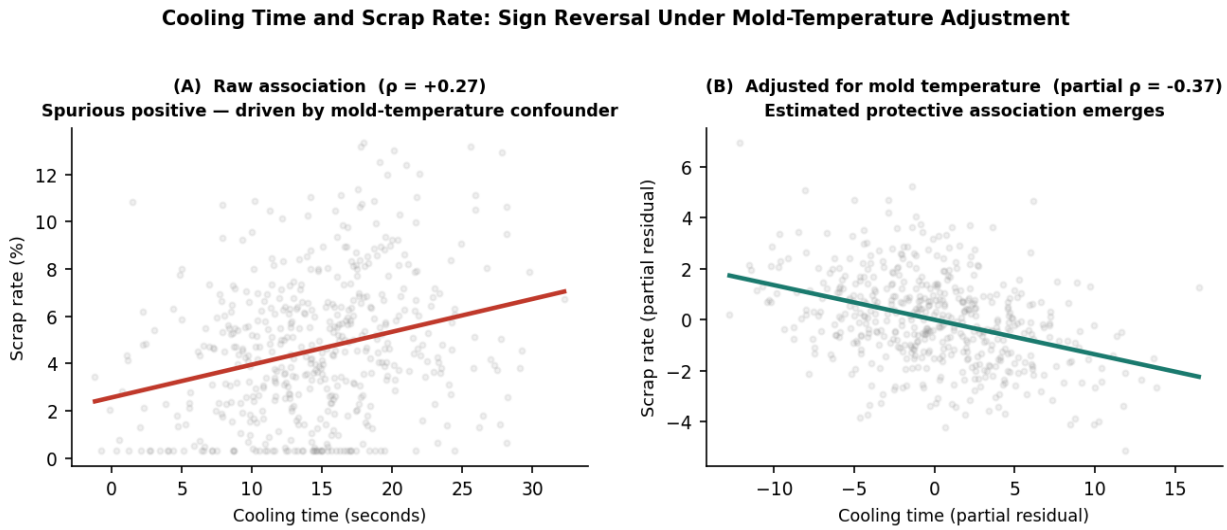

Figure 3 — Illustrative simulation calibrated to the reported summary statistics (raw ρ = +0.28; partial ρ = −0.37); points are not the original production intervals. Panel (A) shows the raw association, where the positive slope is driven by operators compensating for high-mold-temperature conditions. Panel (B) shows partial residuals after conditioning on mold temperature, where the estimated protective association emerges. Actual scatter and density differ from this simulation; the qualitative pattern is consistent with all four estimation methods.

Worked calculation for the +1.5 s intervention (cooling time).

From the natural-unit coefficient (equation 5):

$$\Delta\hat{Y}\ (\text{linear}) = \tilde{\beta_1} \times \Delta t = -0.41\ \text{p.p./s} \times 1.5\ \text{s} = -0.615\ \text{p.p.} \quad (7)$$

The counterfactual simulation (equation 6) refines this for nonlinearity and boundary clipping:

$$\Delta\hat{Y}\ (\text{simulation}) \approx -0.44\ \text{p.p.} \quad (8)$$

The two estimates agree in sign and are close enough in magnitude to support pilot-scale prioritization, though not plant-wide deployment without experimental validation. The simulation value (−0.44 p.p.) is the more conservative figure used in Section 5. The difference between equations (7) and (8) reflects curvature in the GBR response surface near the physical bounds of the cooling-time distribution. All estimates are conditional on the DAG being correctly specified and no material unmeasured confounding.

4.3 Warpage Interaction

To quantify the mold-temperature × cooling-time interaction on warpage scrap, equation (4) is augmented with an interaction term on the warpage subset (n = 1,650):

$$Y_i = \beta_0 + \beta_1 C_i + \beta_2 M_i + \beta_3 (C_i \times M_i) + \gamma^T Z_i + \alpha + \varepsilon_i \quad (9)$$

where C = cooling time and M = mold temperature, both z-scored. A negative β_3 would indicate that the estimated protective effect of additional cooling is amplified at higher mold temperatures.

The cross-tabulation in Table 3 and Figure 4 is directionally consistent with this pattern, while the sparsely populated cells should be interpreted cautiously.

Mold temp × Cooling	≤ 12 s	12–18 s	> 18 s
Mold ≤ 65 °C	3.62% (n=231)	3.81% (n=256)	4.54% (n=17)
Mold 65–75 °C	5.90% (n=20)	4.33% (n=372)	4.44% (n=146)
Mold > 75 °C	8.20%* (n=1)	5.81% (n=110)	5.06% (n=497)

*Table 3 — Warpage-subset mean scrap (%) by mold-temperature × cooling-time bin. * The top-left cell contains only n = 1 interval; this value should be read as directional, not as a stable estimate. The broader pattern — that extended cooling is most beneficial at higher mold temperatures — is consistent across the better-populated cells.*

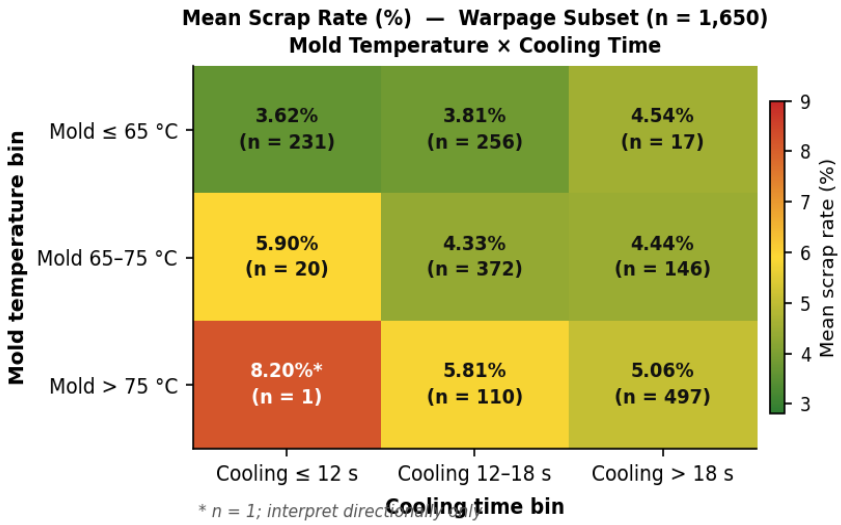

Figure 4 — Mean scrap rate (%) for the warpage subset by mold-temperature bin (rows) and cooling-time bin (columns). Color encodes scrap level (green to red). The asterisked cell (mold > 75 °C, cooling ≤ 12 s) contains only one interval and should not be treated as a stable estimate. The broader gradient confirms that cooling time matters most at elevated mold temperatures.

4.4 Two Effects Disclosed but Not Actioned

Operator experience carries a small positive adjusted β (+0.12). The most defensible interpretation is residual assignment bias: experienced operators tend to be staffed on harder shifts, more demanding product variants, and new tool runs. Shift and variant fixed effects absorb part of this, but the residual positive correlation persists after adjustment. Because this pattern is consistent with assignment bias rather than a direct effect of experience on scrap quality, no operational recommendation follows.

Hold pressure shows a mildly protective adjusted β (≈ −0.09), consistent with process theory that adequate hold pressure reduces sink marks and dimensional deviations. However, the 95% CI narrowly includes zero. The effect is registered in Table 2A and excluded from the primary action package on grounds of borderline significance.

5. Recommendations and Trade-offs

Five coordinated actions on existing equipment are supported for pilot testing. Magnitude estimates are model-implied from chain-propagated GBR counterfactual simulation under the stated DAG assumptions. Each action is rated for evidence strength based on cross-method sign agreement.

5.1 Process Setpoint Actions

Action	Specification	Est. Δ	Trade-off	Feasibility	Confidence
Extend cooling time	+1.5 s plantwide	−0.44 p.p.	+1.0% cycle time; negligible energy	Existing equip.	High
Cap mold temperature	≤ 78 °C	−0.21 p.p.	Negligible	Existing equip.	Med–High
Lower dryer dewpoint	−5 °C when humidity ≥ 65%	−0.09 p.p.	Small energy uptick	Existing equip.	Medium
Reduce injection pressure	−30 bar when wear ≥ 0.45	−0.09 p.p.	Monitor short-shot rate	Existing equip.	Medium

Table 4 — Direct process setpoint actions. Est. Δ = model-implied counterfactual impact in percentage points of scrap rate. These are estimated effects under the stated DAG assumptions, not guaranteed savings.

5.2 Maintenance Scheduling

Action	Specification	Est. Δ	Trade-off	Confidence
Preventive maintenance	Cap interval at 14 days	−0.07 p.p.	Labour cost; no production interruption	Medium

Table 5 — Planning-level action. The reported direct β understates the total effect, which runs primarily through reduced calibration drift. The total-effect estimate is the relevant quantity for operational planning.

5.3 Combined Package Impact

Metric	Baseline	With package (model estimate)
Mean scrap rate	4.44%	≈ 3.85% (−13% relative)
Cycle time	53.7 s	≈ 54.3 s (+1.0%)
Energy per interval	19.01 kWh	≈ 19.02 kWh (+0.07%)

Table 6 — Model-implied combined-package impact. All figures are conditional on the DAG, model specification, and observational data. The dominant trade-off is the cycle-time extension from cooling time (+1.5 s ≈ +1.0%); the scrap-to-cycle-time ratio under the fitted model is approximately 10:1.

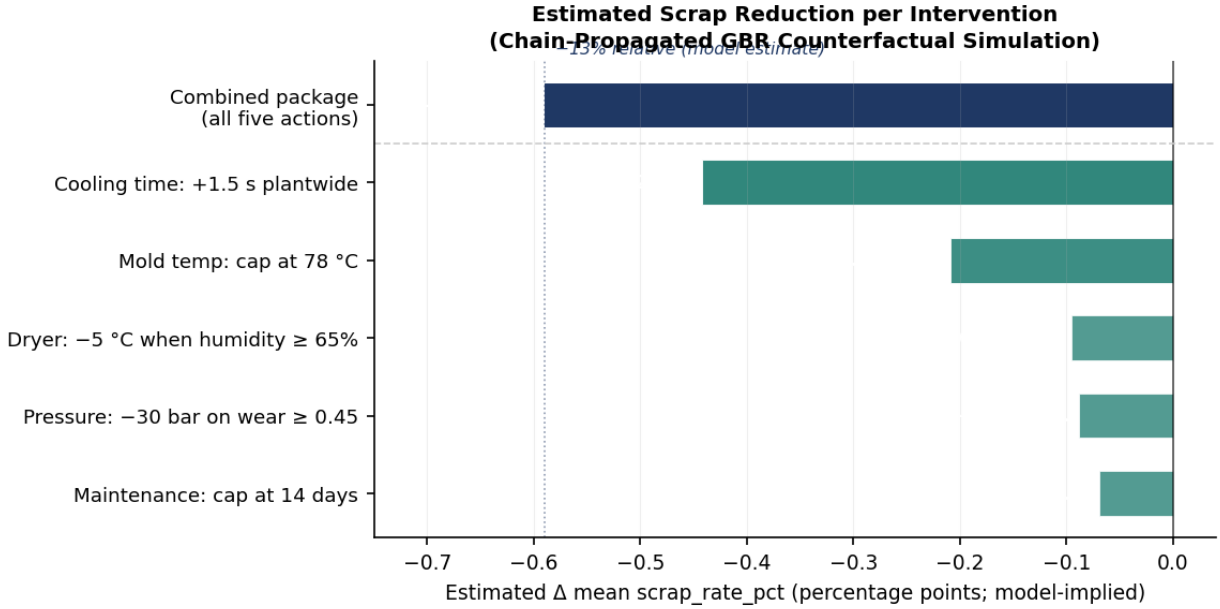

Figure 5 — Model-implied scrap reduction per intervention, sorted by magnitude. Individual action bars (teal); combined package bar (dark blue). The dotted line at −0.59 p.p. marks the combined package estimate (−13% relative vs. baseline). Cooling time alone contributes approximately 75% of the package impact. All values are model-implied estimates, not guaranteed savings.

5.4 Deployment Sequence

Phase 1 (weeks 1–2): Roll out the three lowest-risk setpoint changes — cooling +1.5 s, mold-temperature cap at 78 °C, dryer trigger rule (−5 °C dewpoint when humidity ≥ 65%) — and the 14-day maintenance cap as a scheduling policy change. Pilot at VN_QUANGNAM, which has the highest current scrap rate and the highest humidity exposure. Model-implied target: mean scrap from 4.44% toward ~3.95%.

Phase 2 (weeks 3–8): Introduce conditional pressure control — reduce injection pressure by 30 bar when tool wear index ≥ 0.45. Use short-shot defect rate as a safety gate: if short-shot incidence rises more than 15% relative over any two-week window, roll back and investigate. Model-implied target: mean scrap from ~3.95% toward ~3.85%.

Verification: Track weekly scrap rate by plant and machine. Review defect-mix composition biweekly. Target pass-fail rate below 55% at one month, below 50% at two months.

5.5 Actions Explicitly Not Recommended

- **Shortening cooling time.** The raw positive correlation might suggest this; the DAG-adjusted analysis indicates the opposite. Shortening cooling time would likely increase scrap under the stated model.
- **Plant-wide HVAC to control ambient humidity.** Humidity is a confounder in the ontology — an environmental condition outside the intervention scope. The appropriate lever is the dryer dewpoint.
- **Reassigning experienced operators.** The positive coefficient on operator experience is most consistent with residual assignment bias. No direct causal recommendation follows.
- **Blanket injection-pressure reductions.** On lightly worn tooling, elevated pressure is not associated with elevated scrap. The recommendation is strictly conditional on high wear state (index ≥ 0.45).

6. Limitations

The analysis produces observational decision-support, not experimental proof. All causal effect estimates are conditional on the supplied DAG being correctly specified, no material unmeasured confounding, correct model specification, and the observational — rather than randomized — nature of the data. The following limitations are material to interpreting the quantitative estimates.

Observational identification. Regression adjustment can reduce but cannot eliminate confounding from unobserved sources. Time-varying factors not encoded in the DAG — such as undocumented mold changes, raw-material handling differences across batches, or production schedule pressure — remain potential sources of bias. Fixed effects absorb stable machine-, mold-, and shift-level heterogeneity, but they are not a substitute for randomization. Effect estimates should be treated as plausible operational hypotheses, not proven causal quantities.

DAG uncertainty. The identification strategy depends on the DAG being correctly specified. If there are unmeasured common causes not encoded in the graph, the adjustment set is incomplete and β_1 is biased in an unknown direction. Sensitivity analyses varying the adjustment set confirm that the sign of the cooling-time effect is robust across plausible alternatives; the magnitude is not.

Weak moisture model. The structural sub-model linking dryer dewpoint to resin moisture explains only $R^2 = 0.09$ of moisture variance at the interval level. The three measured parents — ambient

humidity, dewpoint setting, and ambient temperature — are insufficient predictors, possibly because dryer mechanical health, resin storage time, and lot variation are unobserved. The dryer recommendation is sized conservatively and should be treated as a lower-bound estimate. A pilot may reveal greater benefit if the unobserved factors are addressed.

Bootstrap inference. Confidence intervals use a row-wise (i.i.d.) bootstrap. Because adjacent 30-minute intervals on the same machine share equipment-level noise, within-machine observations are not independent. A machine-clustered or time-block bootstrap would produce wider intervals than those reported, more accurately reflecting uncertainty in the magnitude estimates. Signs and rankings are unlikely to reverse under clustering, but magnitudes should be read with corresponding caution.

Temporal and geographic scope. The dataset covers 75 days in winter and early spring. Ambient humidity at the Vietnamese plants in summer is likely to differ, and the humidity-conditional dryer rule is sized on in-sample humidity ranges only. Recalibration for other seasons or plants may be necessary. No external validity claim is made beyond the Karcher group in this time period.

No cost data. All estimates are expressed in scrap percentage points and cycle-time seconds. No monetary conversion is provided. Any financial figures in accompanying materials are illustrative and must be replaced with plant-specific cost data before an investment decision is made.

The two-week pilot at VN_QUANGNAM with defect-mix monitoring as a safety gate is the necessary step to convert these model-implied estimates into empirical evidence and to limit the operational risk of acting on residual confounding.

7. Conclusion

Across 5,000 production intervals, scrap is chronic rather than exceptional: 78% of intervals fail the 3.2% threshold. The dominant source is warpage, driven by insufficient cooling relative to mold temperature and accounting for one-third of all intervals. Flash and dimensional deviation contribute a further 37%, both sensitive to the interaction between tool wear state and pressure settings. Splay moisture has the highest per-interval mean scrap rate and is partially addressable through the dryer dewpoint lever, though the moisture lever's expected impact is bounded by the weak structural model ($R^2 = 0.09$).

The central analytical finding is a sign change for the most important lever. The raw positive correlation between cooling time and scrap ($\rho = +0.28$) reflects reverse causation — operators compensating for thermal stress in high-temperature molds — rather than a direct effect of longer cooling on more scrap. The DAG-adjusted estimate reverses to $\beta = -1.75$ standardized (−0.41 p.p./s natural-unit), indicating an estimated protective association. Acting on the raw signal would likely lead to the wrong intervention. This example illustrates why the causal structure encoded in the supplied DAG is operationally consequential and not methodological scaffolding.

Five coordinated actions — cooling time +1.5 s, mold temperature capped at 78 °C, dryer dewpoint −5 °C under high humidity, injection pressure −30 bar on heavily worn tooling, and maintenance intervals capped at 14 days — are estimated to reduce mean scrap from 4.44% to approximately 3.85% (−13% relative), at a cycle-time cost of approximately 1% and negligible energy impact. All estimates are model-implied from observational data under the stated DAG assumptions. A two-week controlled pilot at VN_QUANGNAM, with defect-mix composition as a safety gate, is the recommended next step.

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